FP Experiment: Rubidium MOT

Tutorial

(Dated: January 2014)

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I. INTRODUCTION

The magneto optical trap (MOT) is an important tool in the wider world of atomic physics. By combining radiation pressure with a quadropole magnetic field, atoms can be slowed down, i.e. cooled, and caught at a confined spot in space. Since 1987, when Steven Chu and his co-workers caught and cooled sodium atoms in a MOT for the first time [1], the MOT has led to fundamental discoveries and practical applications in diverse fields such as quantum degenerate gases (e.g. Bose Einstein condensates), cold collisions, quantum information processing, ultraprecise frequency standards and quantum optics. Therefore, it is not a big surprise that in 1997 Chu was awarded the Nobel price. In Bonn, several MOTs exist in several setups in the experimental groups of the Institute for Applied Physics (Institut fuer Angewandte Physik, IAP).

In this experiment, you will set up a MOT. You will start with tuning the lasers to the right frequencies and adjusting the laser beams before detecting the MOT. Finally you will characterize the MOT. The loading behaviour, the atom number and the dependance on the laser frequency can be investigated.

The Basics section gives you the theoretical background you need to handle the experiment. The MOT setup is not trivial. Knowledge of all experimental parts is crucial for a proper understanding. Elsewise, due to the large amount of free parameters, setting up the MOT can be like looking for a needle in a haystack. For carrying out the experiment, there is a step by step guide helping you with the several experimental parts.

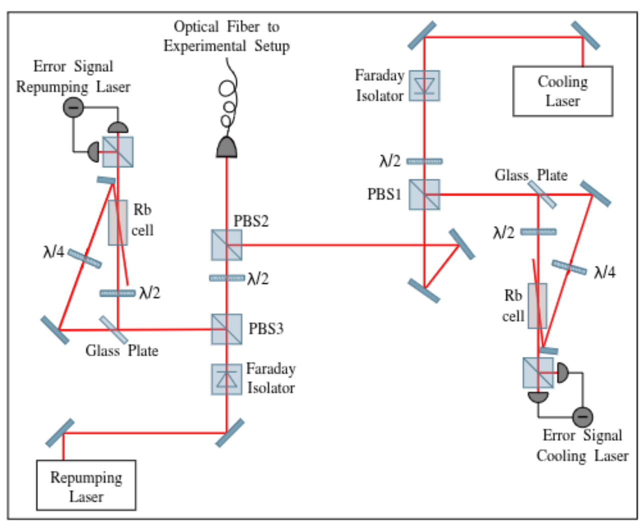

FIG. 1. The laser system part of the experimental setup. The laser frequencies are stabilized via polarization spectroscopy and coupled into an optical fibre guiding the light to the MOT setup. PBS is the acronym for polarizing beam splitter

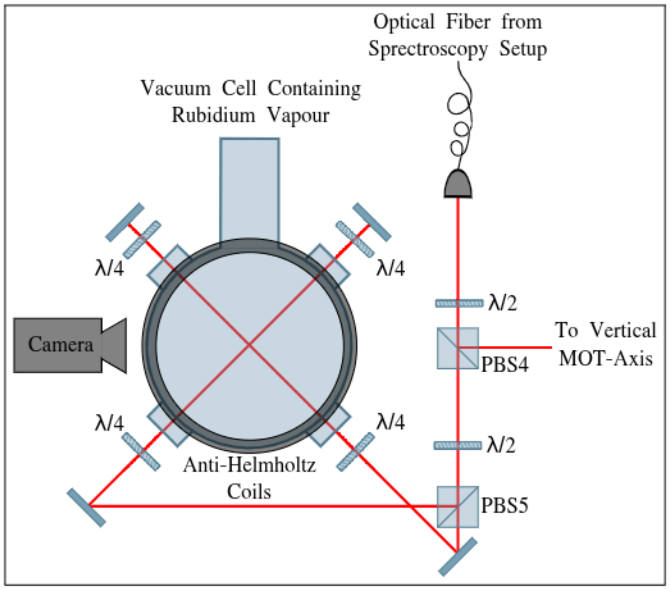

FIG. 2. The MOT setup. The light coming out of the fibre gets split into three beams which enter the vacuum cell from the three different directions of space.

II. BASICS

A. Experimental setup

To give you an overview, we begin with the experimental setup, which is split into a laser system and a MOT part. In the laser system part, to be seen in Fig.1, the beams of the two external cavity lasers are each split into two parts. A small intensity beam is used for spectroscopy in a rubidium cell. By that, the laser frequency can be set and stabilized on both the cooling and the repumping transition. The major part of the laser intensities are overlapped in a polarizing beam splitter (PBS) and then coupled together into an optical fiber. The fiber guides the light to the actual MOT setup (Fig.2). For the MOT, the light is split into three parts, each entering the vacuum cell from a different spatial direction. The vacuum cell, filled with a low pressure of rubidium atoms, is surrounded by a set of coils in anti-Helmholtz configuration which produces a quadrupole magnetic field. When the lasers work at the right frequencies, and the beams are correctly aligned, a MOT can be observed.

B. Interference-filter-stabilized diode laser

The lasers used in the setup consist basically of a laser diode chip, a collimating lens and an partially reflecting window building up an external cavity. An angle dependent filter is placed in the middle of the cavity for wavelength selection. All components are mounted on a solid block of aluminium/brass, as to be seen in Fig.3.

When the laser diode (DL) is driven with a current above the lasing threshold, a divergent light beam ($\approx 15°$) is collimated to a 'proper' beam by a lens (LC). The angle dependent interference filter (FI) assures a correct wavelength selection and allows the usage of an linear cavity design. The cavity is formed with a partially reflective window (OC) and a cat's eye formed by to lenses (L1 and L2) in order to assure a well collimated outgoing beam. By this means, the broad lasing spectrum of the diode chip ($\Delta\lambda >$ mm) can be reduced to a single mode. This diode laser setup is commonly used in atomic physics and was first developed by X. Baillard et al.[3].

Since the length of the resonator cavity determines the laser frequency, slightly varying it with a piezo element results in a shift in frequency. Thus, by applying a triangular voltage to the piezo, the frequency can be scanned over several GHz.

Moreover, the laser frequency depends on both the diode's current and temperature. The laser diode is within the laser cavity, and when it changes its refractive index due to changes in the temperature or electron density, the frequency mode within the cavity is shifted as well.

In summary, we have a single frequency laser whose frequency can be changed by the temperature, the laser current and the cavity length controlling piezo voltage.

C. Spectroscopy

To resolve an absorption spectrum, the laser beam, whose frequency is scanned by ramping the piezo voltage, is guided through a spectroscopy cell filled with rubidium gas and then detected on a photodiode. The atomic gas will be at room temperature, i.e. the atoms move quite fast leading to Doppler-broadening of the spectrum. Fortunately, Doppler-free spectroscopy techniques have been developed to overcome this problem. Further reading in [4].

1. *Saturation spectroscopy*

For saturation spectroscopy, the laser beam is split into a powerful pump and a less powerful test beam, which pass the rubidium cell in opposite directions. The test beam's intensity is then measured with a photodiode. For most frequencies, the spectrum looks like the Doppler-broadened one, because both beams interact with different velocity classes of atoms due to the Doppler shift.

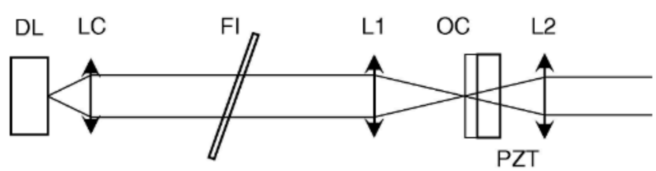

FIG. 3. Basic setup of the diode laser [3].

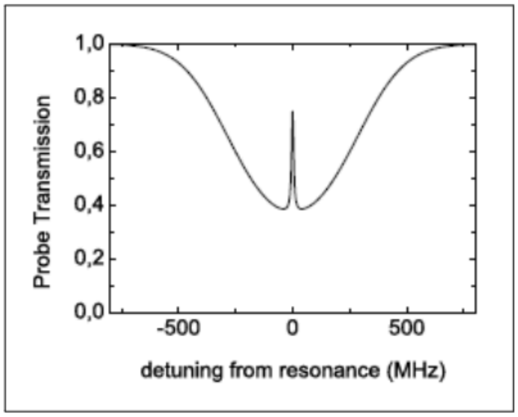

FIG. 4. A typical transmission signal of saturation spectroscopy: In the middle of the Doppler-broadened absorption peak is a Doppler-free peak of higher transmission, called Lamb dip.

Only atoms not moving along the beam axis can be resonant with both beams simultaniously. In that case, the pump beam 'pumps' lots of the atoms in the upper state, leaving less atoms absorbing light of the test beam behind. This higher transmission peak, the so-called Lamb dip, can be seen in the absorption signal of the test beam (Fig.4). Furthermore, it is possible that atoms moving along the beam axis are resonant to a certain hyper fine transition for the pump beam while they are also resonant with the test beam on a different hyperfine transition. As the pump beam is still draining the ground state of the test beam transition, this leads to a high intensity peak in the absorption spectrum as well. These are called crossover peaks.

2. Polarisation spectroscopy

The polarisation spectroscopy technique uses the principles of saturation spectroscopy in an advanced way. While before, the laser was linearly polarized, in this case, the pump beam is circularily polarized after passing a quarter waveplate. Thus, the pump beam will drive the $\sigma^{\pm}$ transitions, resulting to an anisotropic population of the degenerated m_F states. This results in a changing refractive index along one axis in space and thus to birefringence of the material. The test beam, still linearly polarized, is tilted in its polarisation axis by passing the atoms. To resolve the change of polarisation, two polarisators, one before and one behind the cell turned orthogonally, can be used. When the polarisation is not changed, no light passes through the polarisators. Only if it is shifted, light might be detected. In our setup, the test beam passes a half waveplate and is then split in a PBS in two parts and both are detected. Both signals show the normal saturation spectroscopy signature, as the polarisation axis shift is not detected. But substracting the two signals which should be similiar for non-changing polarisation orientiation resolves the axis shift. The polarisation spectroscopy signal is proportional to the change in the refractive index, which is related to the absorption profile via the Kramers-Kronig relation. The result is a dispersive signal looking like the derivative of the Doppler-free absorption peak. Such a signal is perfect for locking a laser.

D. Frequency stabilisation

Frequency stabilisation can be reached by feeding back an error signal to the piezo in the diode laser. A good error signal is proportional to the frequency, i.e. rises and falls with the frequency. Such a signal can be created by polarisation spectroscopy. By adding an offset voltage to the spectroscopy signal, the zero point can be chosen. Via a control box, where you can change the gain, add the offset voltage etc., the signal is applied to the piezo controlling the cavity length. When the laser drifts in its frequency (vibrations, temperature changes, etc.), the error signal rises and the cavity legnth will be changed to counteract the frequency drift.

E. Optical cooling

Photons have a momentum of $p = \hbar k$. Whenever a photon gets absorbed by an atom, it will transfer this momentum to the atom. On the other hand, photon emission will give a recoil momentum to the atom. If a laser beam is guided onto an atom, i.e. lots of photons 'hit' the atom from one side, the atom will feel a force $F = dp/dt$ from the photon momentums. In this case, the recoil force can be neglected, as the photon emission occurs isotropically in every direction of space and accordingly sums up to zero for a large number of photons.

Due to the Doppler shift, the atom's resonance with the laserbeam depends on their velocity in beam direction. If you detune your laser to the red, only oncoming atoms will feel the radiation pressure and thus slow down. By adjusting the resonance by changing the laser frequency or manipulating the atom resonance (Zeeman effect), the atoms can be slowed down effectively. By cooling the atoms from all directions in space, the average velocity and hence the temperature of the atomic gas can be cooled.

This cooling process is limited by the recoil process. The cooled atoms perform a random walk driven by the momentum of the emitted photons, which heats the atoms. The so called Doppler limit is reached when heating and cooling rate cancel each other out.

F. Magneto optical trap

By setting up six red detuned laser beams, two counter-propergating ones in each of the three directions of space, the atoms can be cooled and form an 'optical moloasses'. Nethertheless, these cooled atoms can move in the atom cloud and leave the intersection volume of the beams. This problem is solved in the MOT by combining the cooling force with a force dependant on the position in space using the Zeeman effect. By applying a quadrupole magnetic field using two coils in anti-Helmholtz configuration, with its zero point in the middle of the intersection volume of the beams, a linear magnetic field is observed in every beam's direction. In the middle of the so formed magneto optical trap (MOT), the cooling forces act as before (B=0). But if an atom drifts away from the middle, its resonance gets shifted by the magnetic field towards the frequency of the red detuned laser beams. The rising force will push the atom back to the middle of the MOT. To pick the right Zeeman-shifted transition, the laser beams are circularily polarized.

G. Rubidium MOT

Rubidium exists in two stable isotopes, ^{85}Rb and ^{87}Rb. In this experiment, the ^{85}Rb isotope is caught in the MOT. Nethertheless, ^{87}Rb atoms are in both the vacuum chambre and spectroscopy cell, so that their spectrum is detected. The ^{85}Rb level scheme is shown in Fig.5. As you can see it is more complex than a simple two level system. According to the σ^+ light in the MOT, we choose a cooling transition with $\Delta F = 1$. We choose the F=3 $\rightarrow$ F'=4 transition, because this transition is nominally closed, meaning that atoms will continue to scatter laser light through many absorption/emission cycles because the atom in the exited state can only decay back into the ground state of the transmission. However, rare off-resonant excitation to the F'=2,3 excited states does allow the atom to decay to the F = 2 ground state, where it is far-detuned from laser light and thus lost to the laser cooling process. To mend this problem, we introduce laser light resonant with the F=2 $\rightarrow$ F'=3 transition, pumping the atoms back into the cooling transition when the atom decays into the ground state of the cooling transition.

Also read the first three pages of the paper 'Inexpensive laser cooling and trapping experiment for undergraduate laboratories' in the appendix for a more detailed explanation of the laser cooling process, the magneto optical trap and trapping atoms in a Rb vapor cell. Further reading in [5, 6].

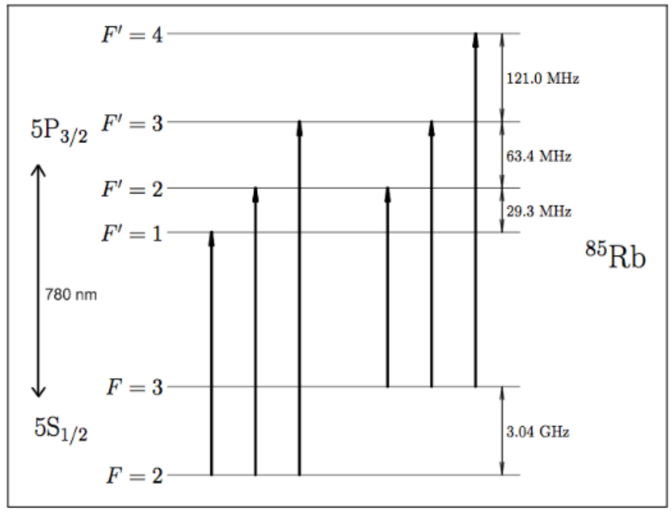

FIG. 5. ^{85}Rb level structure. The F=3 $\rightarrow$ F'=4 transition is used for the cooling beam, while the repumping laser frequency is adjusted to the F=2 $\rightarrow$ F'=3 transition

III. TASKS AND STEP-BY-STEP EXPERIMENT GUIDE

A. Your tasks

First, and most importantly, it is your job to get the MOT running. This includes locking the lasers to the correct frequencies and adjusting the MOT beams. In the second step, you should investigate the MOT with a photodiode/powermeter. By measuring the fluorescence, you can calculate the atom number. Based on that, you can take several measurements testing the influence of frequency detuning, laser polarisation etc. on the MOT population.

It is important to state that this experiment is not easy and it requires some skill to catch atoms in the MOT. The following qualitative measurements should be easier. The amount of measurements you accomplish in the two days of experimentation are related to your mark.

List of tasks:

- Understand the laser system and learn how to handle it properly.
- **Adjust the MOT with the help of the step by step guide.**
- Install a powermeter to measure the MOT fluorescence.
- Determine the MOT atom population.
- Investigate the loading behaviour of the MOT.
- Estimate the MOT size.
- Record the influence of the MOT beam polarisation on the MOT population.

FIG. 6. Control unit of a diode laser. While working on the experiment, you will only change the diode current which you do by turning the knob in the lower right corner. The right display shows the current value.

- Record the atom number as a function of the magnetic field.
- Record the atom number in the MOT while scanning the laser frequency.

B. Be careful!

As the experiment is rather complex, a lot of things in the setup can be destroyed or misaligned by the laboratory worker. Make sure you always know what you are doing, or at least have a good idea. In the setup, there are several parts not to be touched. That is basically everything you're not told to touch in the step by step guide, but includes

- the breakable vacuum cell, including the pumps,
- the whole laser setup,
- and all the items fastened onto the optical table, including all the postholders, the quarter waveplates, the beamsplitters, the camera and the magnetic coils.

When you're not sure what to do, call your tutor!

C. How to use the laser

Both the temperature and the diode current can be set with the laser control units. You should not change the temperature, but the current can be changed with the big knob in the lower right corner. The required current values can differ from time to time, but your tutor should be able to give you an estimated value. For scanning the laser, a frequency genererator creates a triangular signal which is fed to the lock boxes of the lasers. In the lock box, you can change the amplitude of this signal and you can add an offset voltage. The resulting voltage is applied to the controlling piezo of the laser cavity.

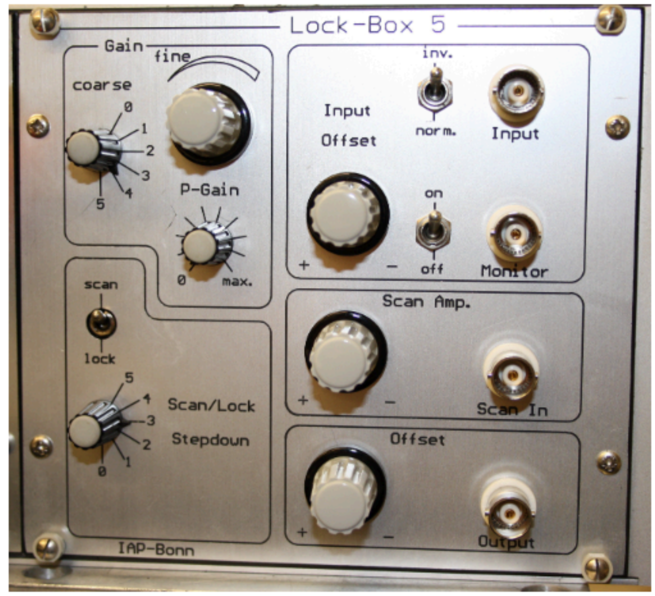

FIG. 7. The LockBox. Start by setting the Scan/Lock switch on Scan. In this mode, the 'Scan In' signal (from the frequency generator) will be fed to the output. It's amplitude can be modified from max to zero, and an offset value can be set.

When the scan amplitude is on maximum, you can search through the laser modes by playing with the offset voltage and the diode current. You should do this to find the rubidium spectrum, which can be seen on the oscilloscope.

D. Find the cooling transitions

When you have found the spectrum, you can carfully tune the laser with scan offset, scan amplitude and laser current. Try to identify the different peaks in the spectrum with the help of the spectrum graphs at the end of this tutorial. Check if you can see fluorescence in the vacuum chamber.

E. Locking the laser

Normally, the lasers are locked onto the atomic transition and the MOT beams are detuned by acustooptic modulators (AOM). To keep the experiment as easy as possible, we don't use AOMs, but adjust the lasers directly to the desired frequency. By reducing the Scan and adjusting the offset the laser can be set to the desired wavelength.

However, the MOT population is quite sensitive to the cooling laser frequency. By playing with the different waveplates in the polarisation spectroscopy we observe an error signal even for frequencies lower than the cooling transition. By turning the input offset knob, the error signal can be set to zero at the desired wavelength. If you switch the Lockbox from scan to lock, the error signal will be fed back to the piezo. When the laser is locked, the signal should oscillate arround zero. When you cause some vibrations, this should be seen in the error signal. By changing the gain, you can reduce the noise in the error signal. Slight changes in the laser frequency can be reached by turning the signal offset voltage.

F. Check power

Firstly, make sure you have enough light passing through the fibre. By turning off the repumping laser, you can measure the beam power of each laser with the powermeter. The cooling beam should have at least 5 mW, while 1 mW is enough for the repumping beam. When the power is too low, the coupling into the fibre has to be readjusted. Before doing that on your own, talk to the tutor so he can show you how to do this.

Furthermore, the cooling beam power in the several MOT axis have to be leveled to equal. To do so, switch off the repumping beam (its power distribution doesn't matter) and change the several beam powers by turning the $\lambda/2$ waveplates in front of the PBS.

G. Adjusting the MOT beams

There are six post holders and six mirrors. Put the mirrors in the postholders to adjust the MOT beams (The mirror with the long post belongs underneath the vacuum chamber). Be careful doing this. Don't touch the setup, even though it is hard for the mirrors under the board. Furthermore, be careful of randomly reflected beams while adjusting the beams. Block the beams until you have roughly adjusted the mirrors. For fine adjustment, obviously, you start with the incoming beam. Make sure it passes both chamber windows in the middle. To see this properly, you can use the little camera, which is, different to the eye, still reasonably sensitive to a wavelength of 780nm, or you can use the infrared viewer. Then, you adjust the backreflection the same way. Make sure that the backreflection overlays the incoming beam all the way back to the fibre.

Then, check your adjustment. You can do this by looking at the fluorescence in the chamber using the viewer/cameras (make sure the laser frequencies are correct!). The beams are meant to perfectly cross in the middle. If they don't, correct this.

H. Find and optimize the MOT

Switch on the MOT coils power supply and put the current on a value of about 1.5 A. Then check your laser frequencies again. When the beams are resonant with the rubidium atoms, you see flourescence in the chamber. Have a look at the CCD camera: Lock the cooling laser to a frequency about a linewidth to the red of the transition. You should see fluorescence in the chamber. By turning the repumping laser frequency with the offset knob, the fluorescence can be optimized. The brightest fluorescence gives you the perfect repumping laser frequency. Now, everything is set to find the MOT. Tune the cooling laser frequency slightly by turning the 'Input Offset' button on the lock box, and see if you can spot a MOT.

It is quite likely you dont see anything directly, because now there are several parameters which all have to be set precisly at the same time to find a MOT. The parameters you are meant to play with are the beam adjustment and the laser frequencies.

When the beams are adjusted as described, the magnetic fields zero point should be in the middle. Netherthe-less, it might have to be changed slightly. As this brings a high risk of unaligning the whole setup, you should contact your tutor to get him to him check your setup before changing the coils position. An easy method to move the zero point of the magnetic field is to apply an offset field with a permanent magnet. You can try this on your own.

After you found a little MOT, optimize it by optimizing the mentioned parameters.

I. Install powermeter

If you got this far: Congratulations, your MOT is running. Now set up a detector to detect the MOT fluorescence. Use a lens to image the MOT onto the powermeter head and measure its fluorescence. Make sure you that you really image the MOT and not a reflection. To check if you detect the MOT fluorescence, you must switch the magnetic field on and off and substract the background. You can't just block the beams as they are responsible for lots of the underground in the chamber. If you get a signal, optimize it by finding the perfect position. Then, you can optimize your MOT with this signal (about 100 nW possible, about 50 nW are needed).

J. Characterize MOT

When the powermeter is measuring an optimized fluorescence signal, you can determine several MOT characteristics.

1. MOT population

From the MOT fluorescence, you can calculate the atom number caught in the MOT. To do so, you will have to measure several parameters. These are the fluorescence value, the lense diameter, the distance between lense and MOT, the detuning (will be determined in part

III J 6 of the experiment) and the beam power and diameter. Most values can be easily measured. When measuring the beam power, make sure the whole beam is detected. The beam diameter can be measured using a razor blade on an optical rail. Record the beam power depending on the blades position and fit this with the errorfunction (the integral over the gaussian beam). Thus you can determine the $1/e^2$ radius of the beam. For further details see [7] section *VI. C. Measurements* and notice that formular (3) from [7] section *II. C. Rb vapor cell trap* is not appropriate here!

2. Size of MOT

Take an image of the MOT, then tilt the camera and put a millimeter scale in the focus to calibrate the camera solution (mm per pixel). Estimate the MOT size with the help of the calibration value.

3. Influence of quarter waveplate

Measure the influence of the quarter waveplate angle on the MOT population using the powermeter. Make sure you optimize the waveplate to maximum MOT population after the measurement. Compare the influence of the waveplates passed by the incoming beam and the ones only passed by the backreflected beam.

4. Changing the magnetic field

Change the magnetic field by tuning the current of the coils and measure the MOT fluorescence using the powermeter. Plot and discuss the obtained data.

5. Loading behavior

Replace the powermeter measuring the MOT fluorescence by a photodiode. Connect the photodiode to the digital oscilloscope and set the timescale on about 0.5 s. After unblocking the beams or switching on the MOT field, observe the loading behaviour of the MOT. Take the measurement several times and derive a mean loading time.

Furthermore, record the pressure in the vacuum cell (the pump controller can be found on the back side of the experiment, under the table), and recorde the room temperature. Since the loading behaviour is determined by the background collision rate (collision rate with background rubidium atoms!!!), the Rb-Rb cross section can be calculated! Therefore, we assume these collisions to be dominant and neglect collisions with other elements. For further details see [7] section *II. C. Rb vapor cell trap.*

6. Detuning of the laser frequencies

Measure the MOT fluorescence as a function of the frequency of the cooling laser while the frequency of the repumping laser is keept constant. After that measure the MOT fluorescence as a function of the frequency of the repumping laser while the cooling laser is locked. For a proper quantitative measurement, you should callibrate your frequency detuning with the spectrum. For the laser whose frequency is scanned, connect the corresponding spectrum signal and the fluorescence signal of the photodiode to the digital oscilloscope. Switch the scan frequency to very slow values ($< 1\,\mathrm{Hz}$), so your measurement is slower then the loading time of the MOT. Like that you get a frequency-dependend measurement of the MOT fluorescence, which you can callibrate by means of the known rubidium spectrum.

IV. EXAMINATION

A. Oral entrance examination

For the sake of being able to carry out the experiment, you should know about the following keywords. Before the experiment, you will be tested on your knowledge.

- **optical cooling:** radiation pressure, red detuning, Doppler shift
- **optical molasses:** counterpropagating beams, 3D, cooling but no caught atoms, Doppler temperature
- **MOT:** Magnetic quadrupole field, circular polarized beams, position-depending force
- **Rb-MOT:** Rb energy level structure, non-resonant excitation, inelastic collisions, dark states, repumping beam
- **setup:** coils, vacuum chamber, laser system
- **Doppler-free spectroscopy:** pump/signal beam, groups of atoms with similar velocity, Lamb dip, crossover resonance, Rb-spectrum
- **polarisation spectroscopy:** circularly polarized light, anisotropic pumping, birefringence, detection of tilt, dispersion, Kramers-Kronig relation
- **diode laser:** tunable frequency, diode current, piezo
- **frequency lock:** error signal, lockbox, offset voltage

B. Labreport

First of all, read the 'report guidelines' applying for all experiments. The following ideas on how to write your report are based on the guidelines.

- **abstract:** Begin with a short abstract describing the main facts: What is the report about and what are the major achievements?
- **theory:** Give a brief introduction to the topic. No long explanation of details, but enough information to explain all the major ideas. Remember: A well-written theory part requires a good understanding by the author.
- **setup:** Include all important parts of the experiment.
- **measurements:** Please structure your report by the several tasks given. For each part, mention the experimental methods, show the data, analyse the data and discuss your results! Don't forget the part about adjusting the MOT, even though there is no data recorded.
- **conclusions:** Sum up your results. If you have useful suggestions to improve the experiment, do not hesitate to write them down!

The evaluation of the labreport will be based on the following criteria: Theory + Setup (10%), section about adjusting the MOT (10%), care taken in measurements and analysis (20%), correctness of measurements and results (20%), discussion of the results (20%), the overall readability which includes a clear structure from start to finish (15%) and finally the scientific standards (5%).

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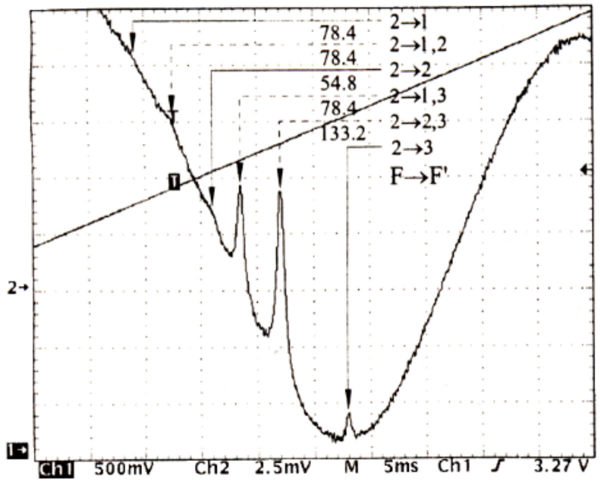
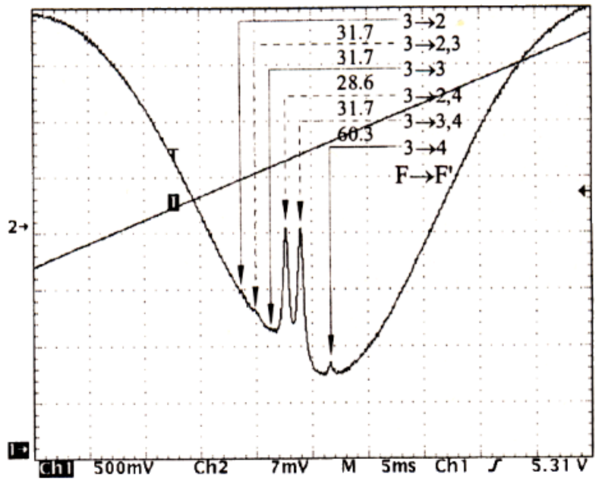
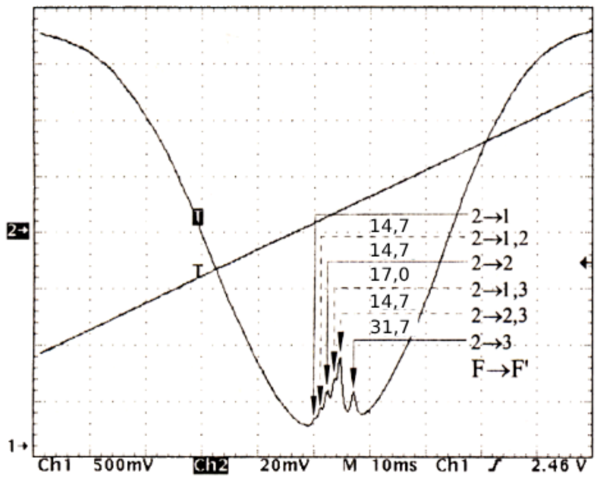


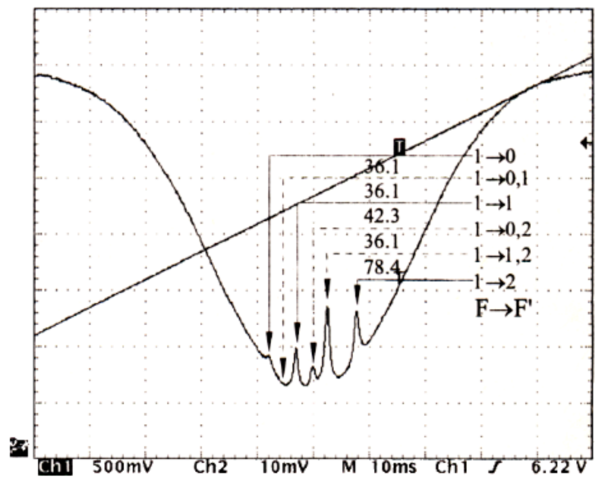
Rubidium 87, $F = 2 \to F' = 1,\ 2,\ 3$



Rubidium 85, $F = 3 \to F' = 2,\ 3,\ 4$



Rubidium 85, $F = 2 \to F' = 1,\ 2,\ 3$



Rubidium 87, $F = 1 \to F' = 0,\ 1,\ 2$